VHDL Assignment 1

Subject: Digital Logic Design

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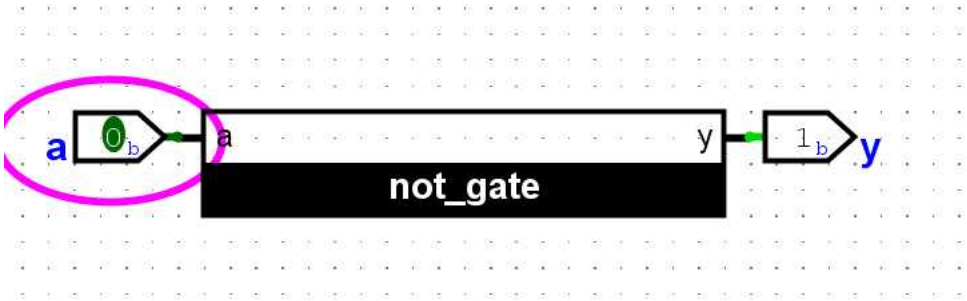
1. VHDL source code

```vhdl
library ieee;
use ieee.std_logic_1164.all;

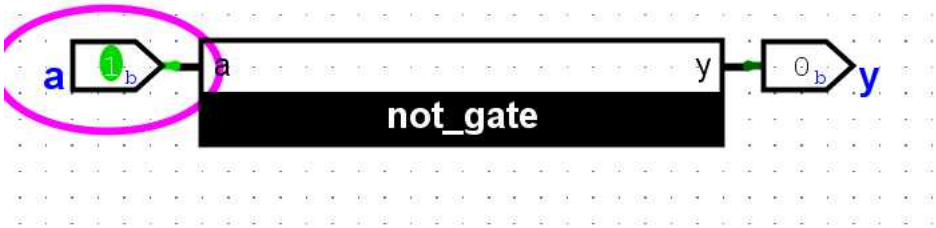
entity not_gate is
  port (
    a : in  std_logic;
    y : out std_logic
  );
end entity not_gate;

architecture behavioral of not_gate is
begin
  y <= not a;
end architecture behavioral;
```

2. Logisim project



- case 1) a = 0 → y = 1



- case 2) a = 1 → y = 0

3. Result explanation

not_gate is a logic gate that inverts its input.
When the input a is 0, the output y becomes 1, and when the input a is 1, the output y becomes 0.

a	y
0	1
1	0

[Table 1] Result truth table

As shown in the truth table, the output y is the inverse of the input a.

Therefore, the VHDL simulation of the NOT gate works correctly.